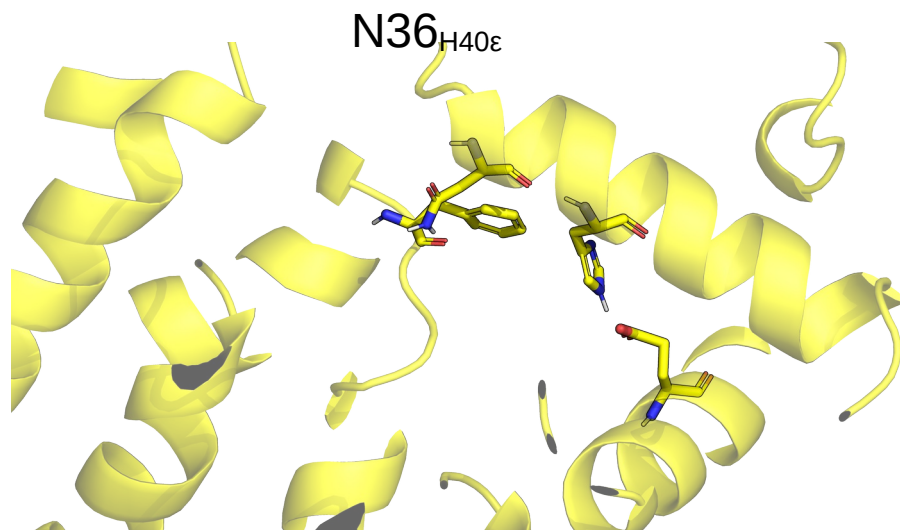
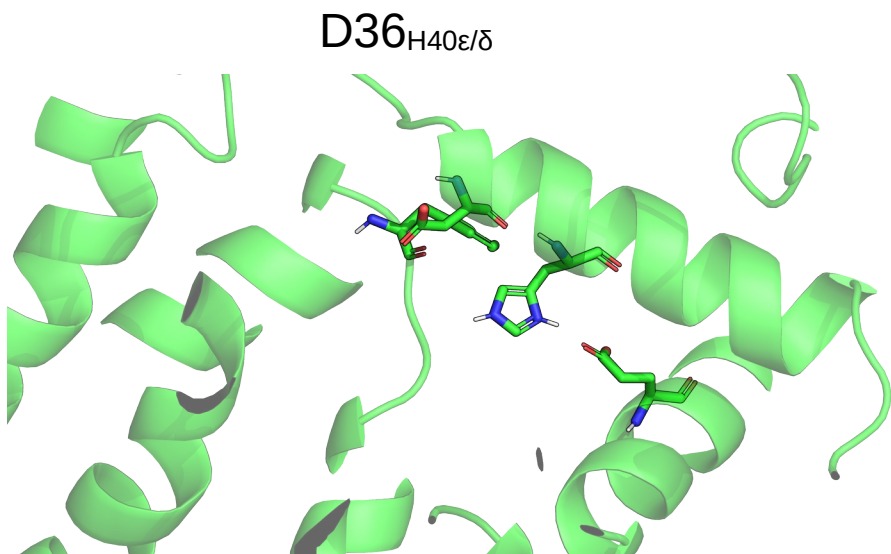
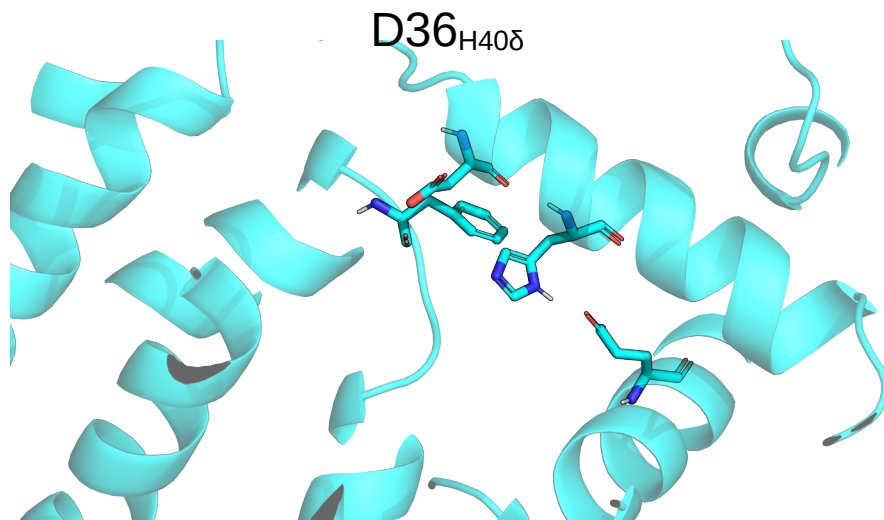
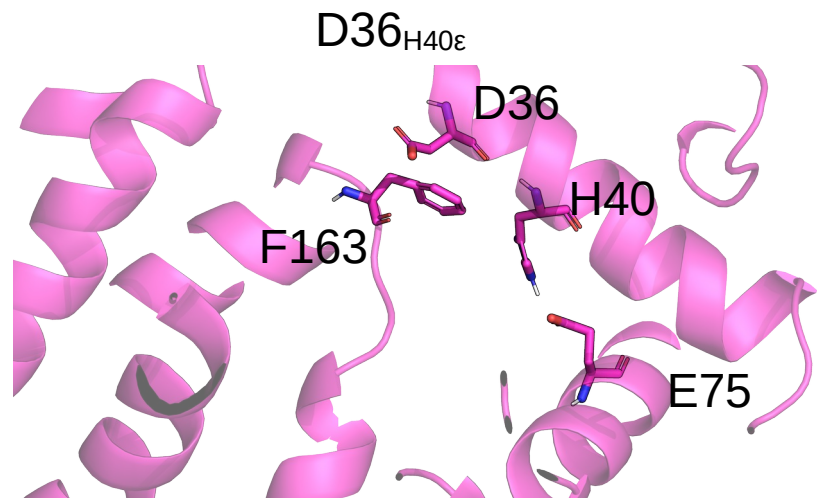
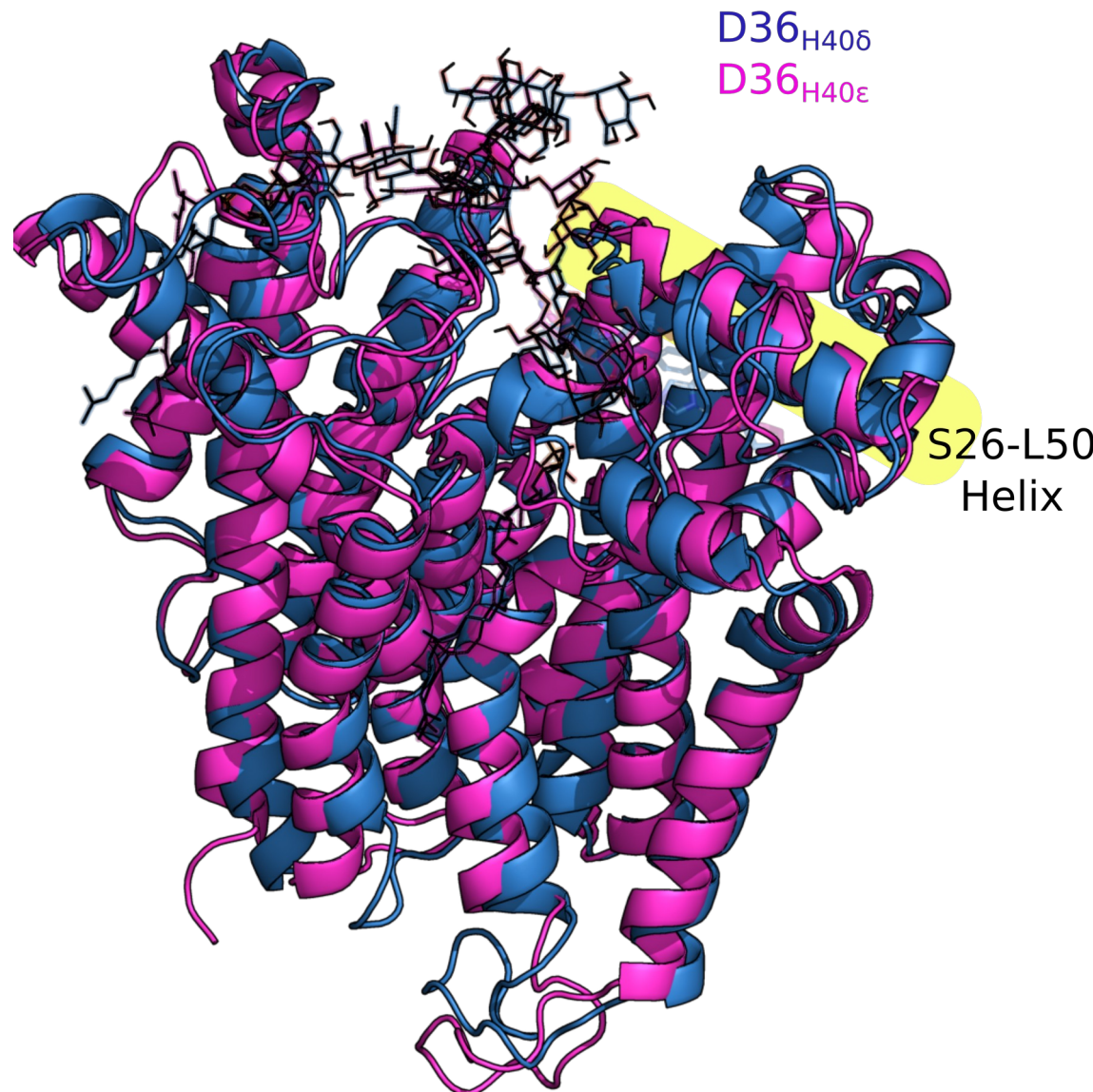
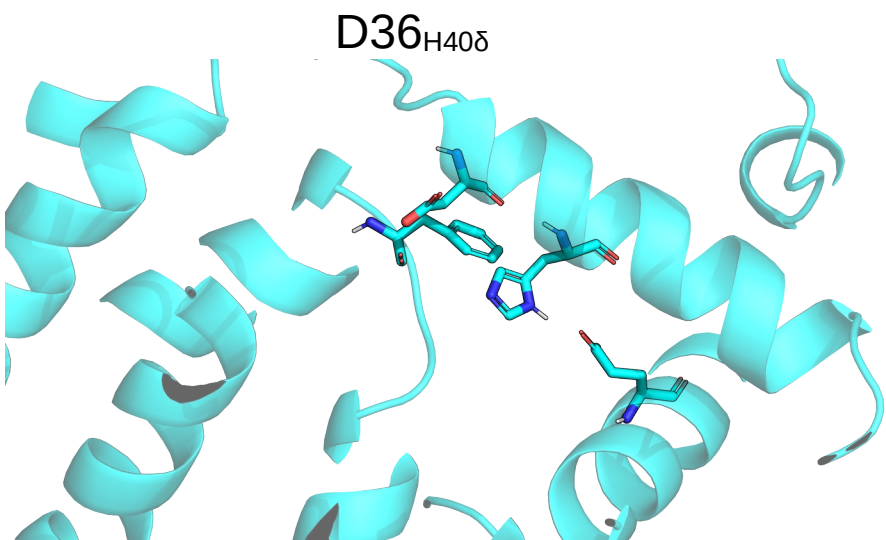
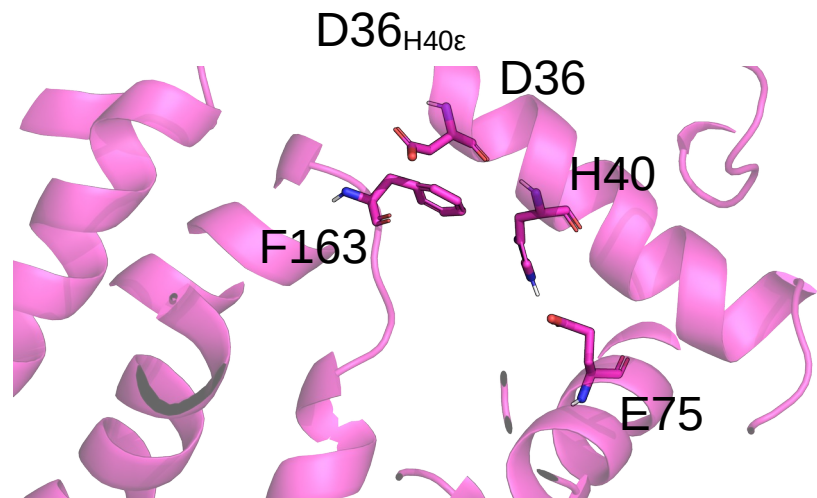


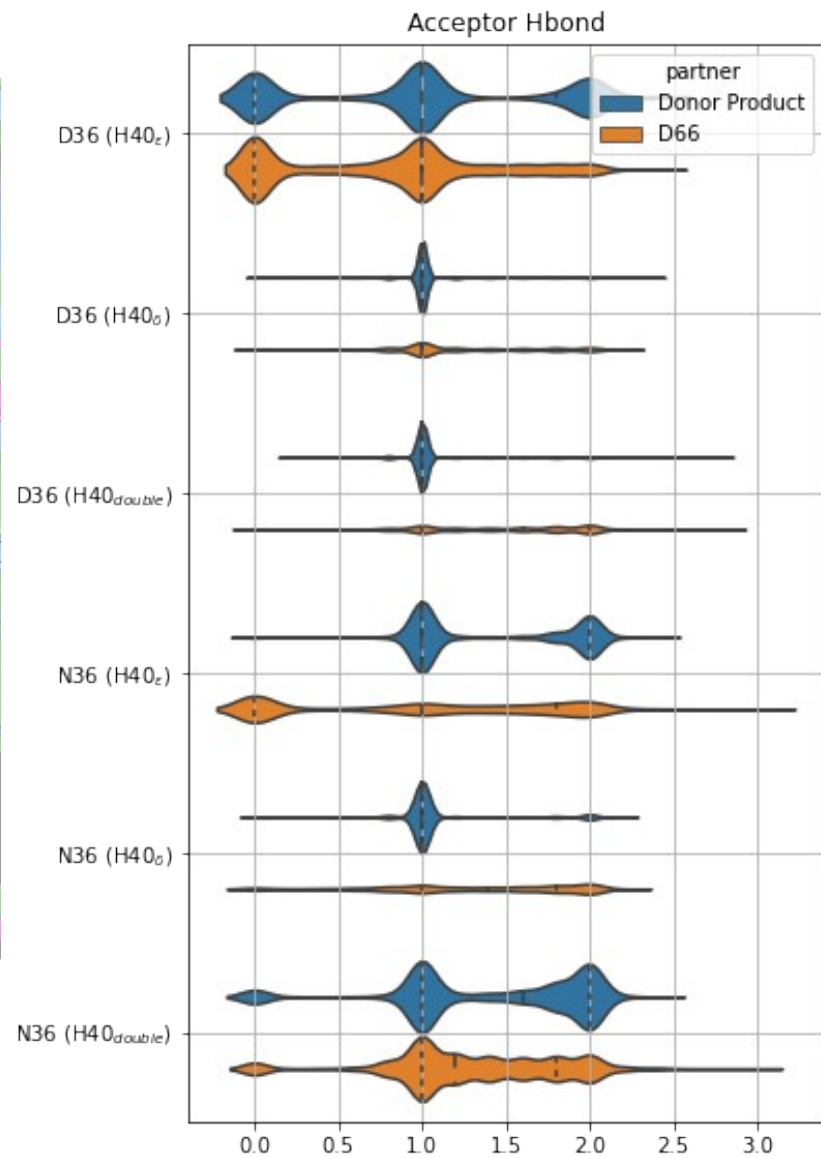
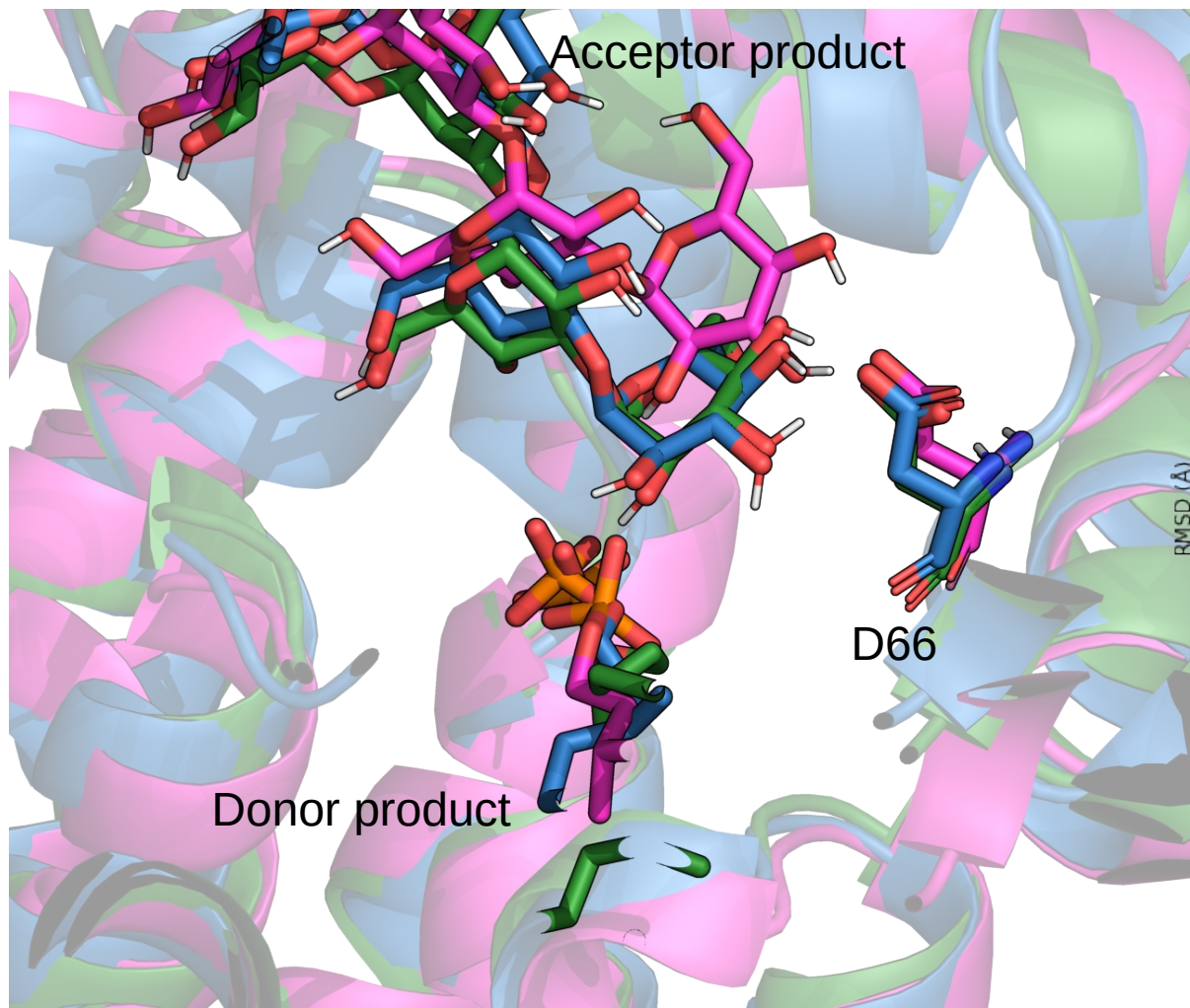
Chapter 1

Posttransfer:

Why D36N does not release the substrate

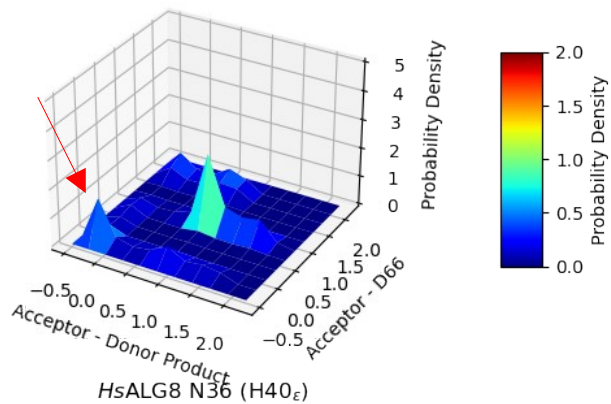




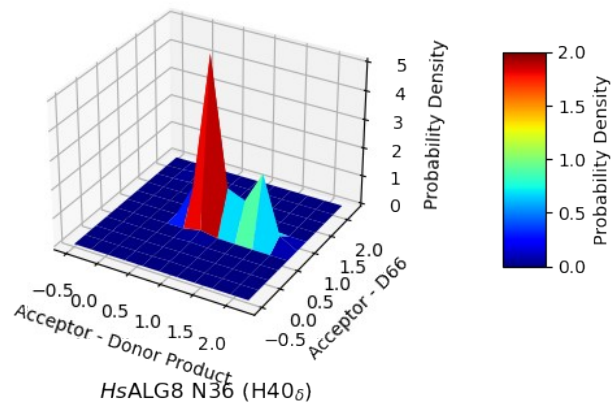


H-bond counts

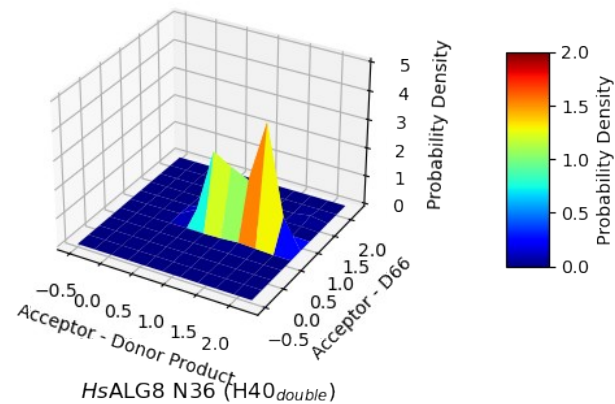
*Hs*ALG8 D36 ($H40_{\epsilon}$)



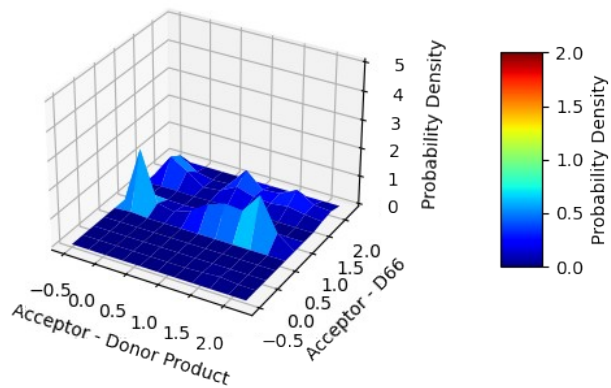
*Hs*ALG8 D36 ($H40_{\delta}$)



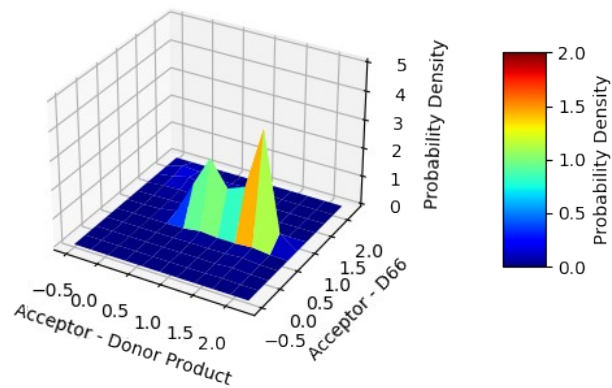
*Hs*ALG8 D36 ($H40_{double}$)



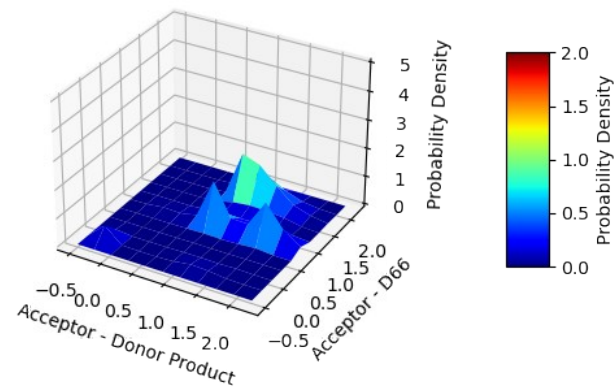
*Hs*ALG8 N36 ($H40_{\epsilon}$)

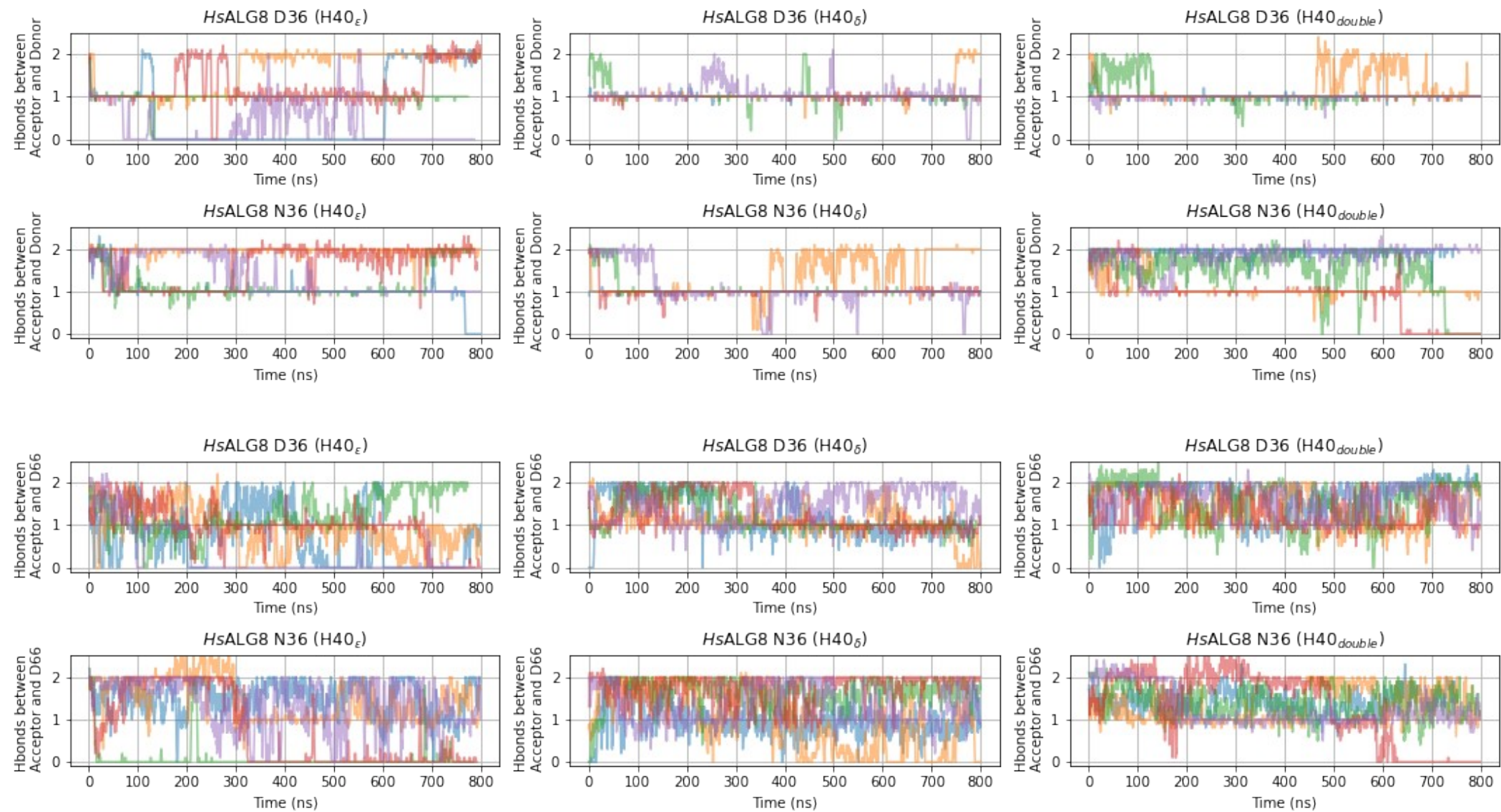


*Hs*ALG8 N36 ($H40_{\delta}$)

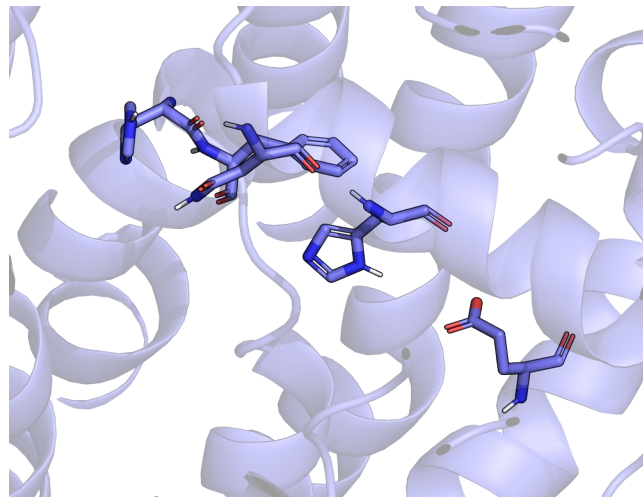


*Hs*ALG8 N36 ($H40_{double}$)

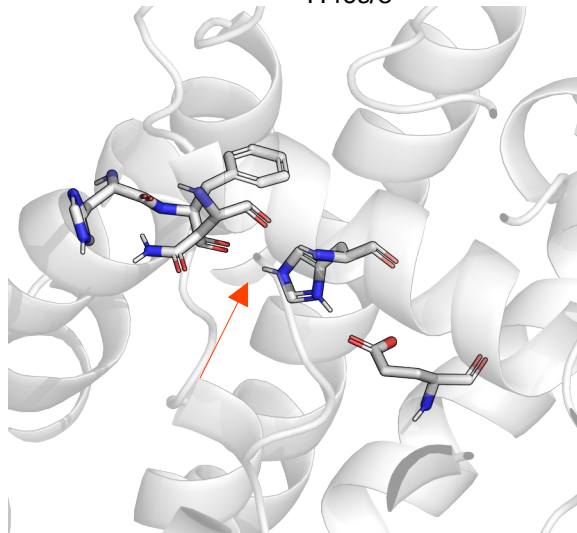




N36_{H40δ}

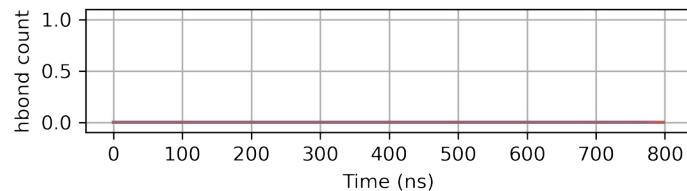


N36_{H40ε/δ}

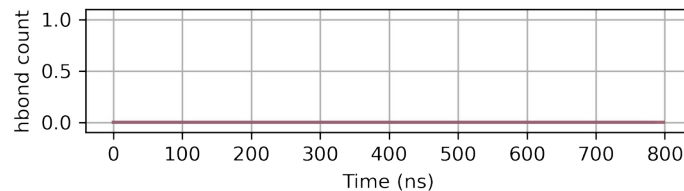


In this case, H40 forms H-bond with F163, N36 forms H-bond with H162 instead.

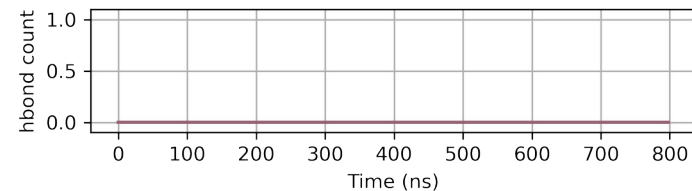
*Hs*ALG8 D36 (H40_ε)
D/N36 - F163 Interaction



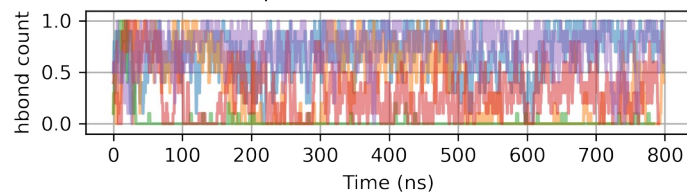
*Hs*ALG8 D36 (H40_δ)
D/N36 - F163 Interaction



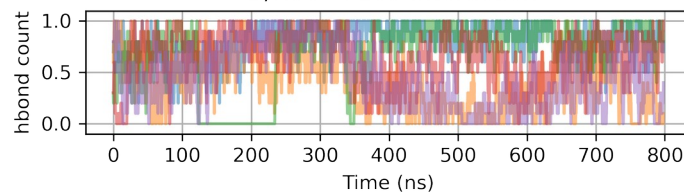
*Hs*ALG8 D36 (H40_{double})
D/N36 - F163 Interaction



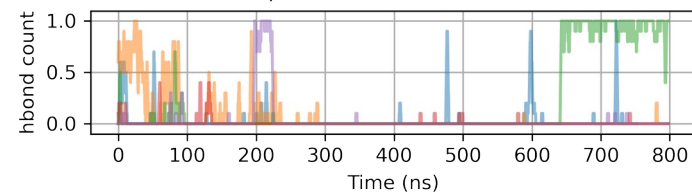
*Hs*ALG8 N36 (H40_ε)
D/N36 - F163 Interaction



*Hs*ALG8 N36 (H40_δ)
D/N36 - F163 Interaction



*Hs*ALG8 N36 (H40_{double})
D/N36 - F163 Interaction



Summary of Chapter 1

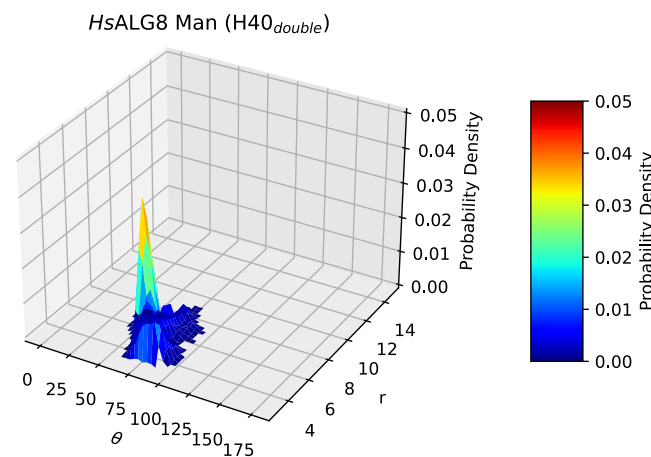
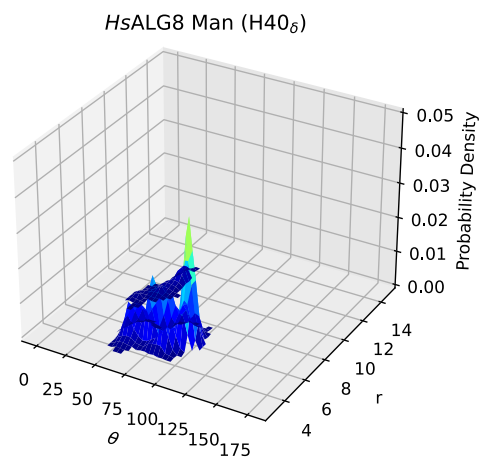
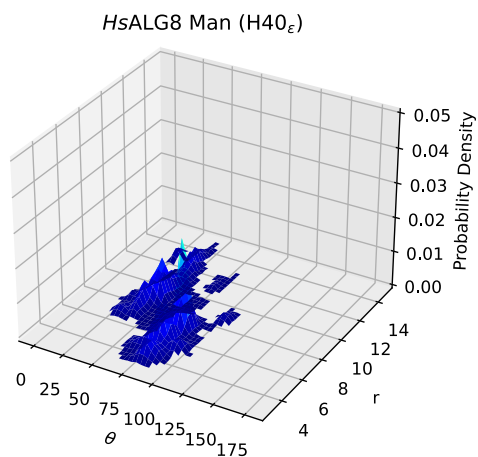
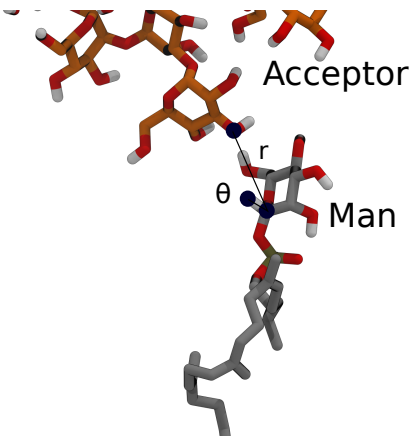
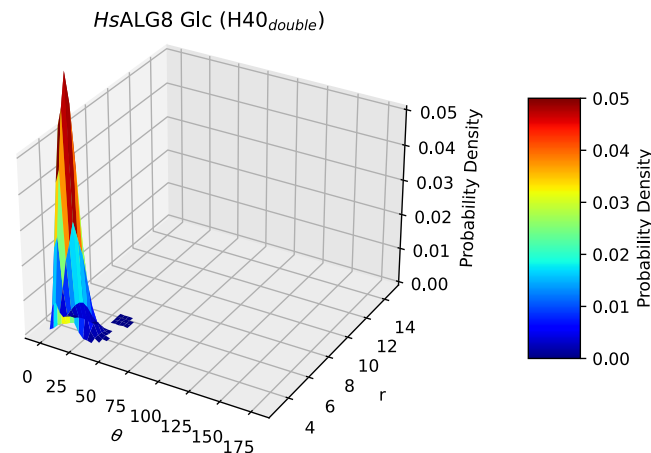
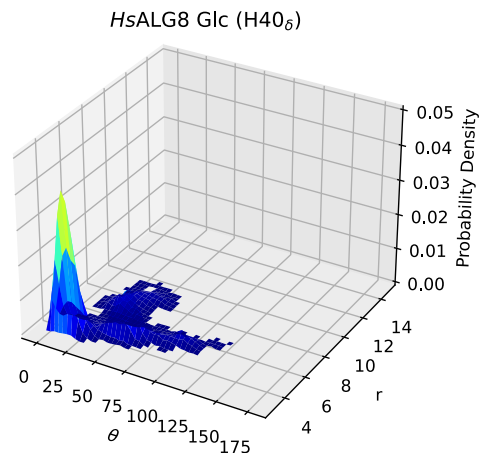
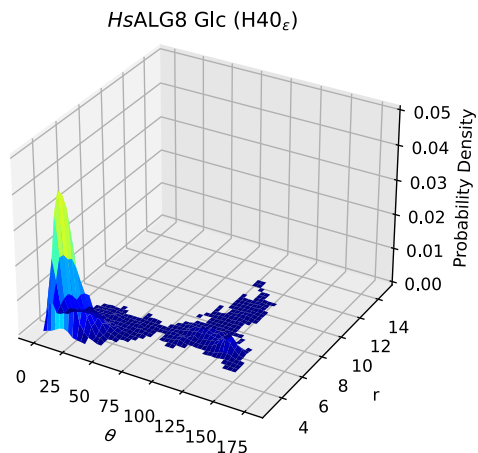
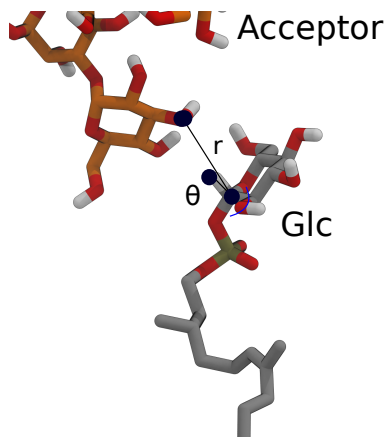
- The protonation state of H40 seems to be very crucial for the flexibility of the enzyme, especially the S24-L50 helix. In the D36_{H40ε} simulations, we observe helix opening and substrate product leaves the active site (losing hbond with donor and D66)
- In D36N, this helix remains closed because of the extra N36-F163 Hbond

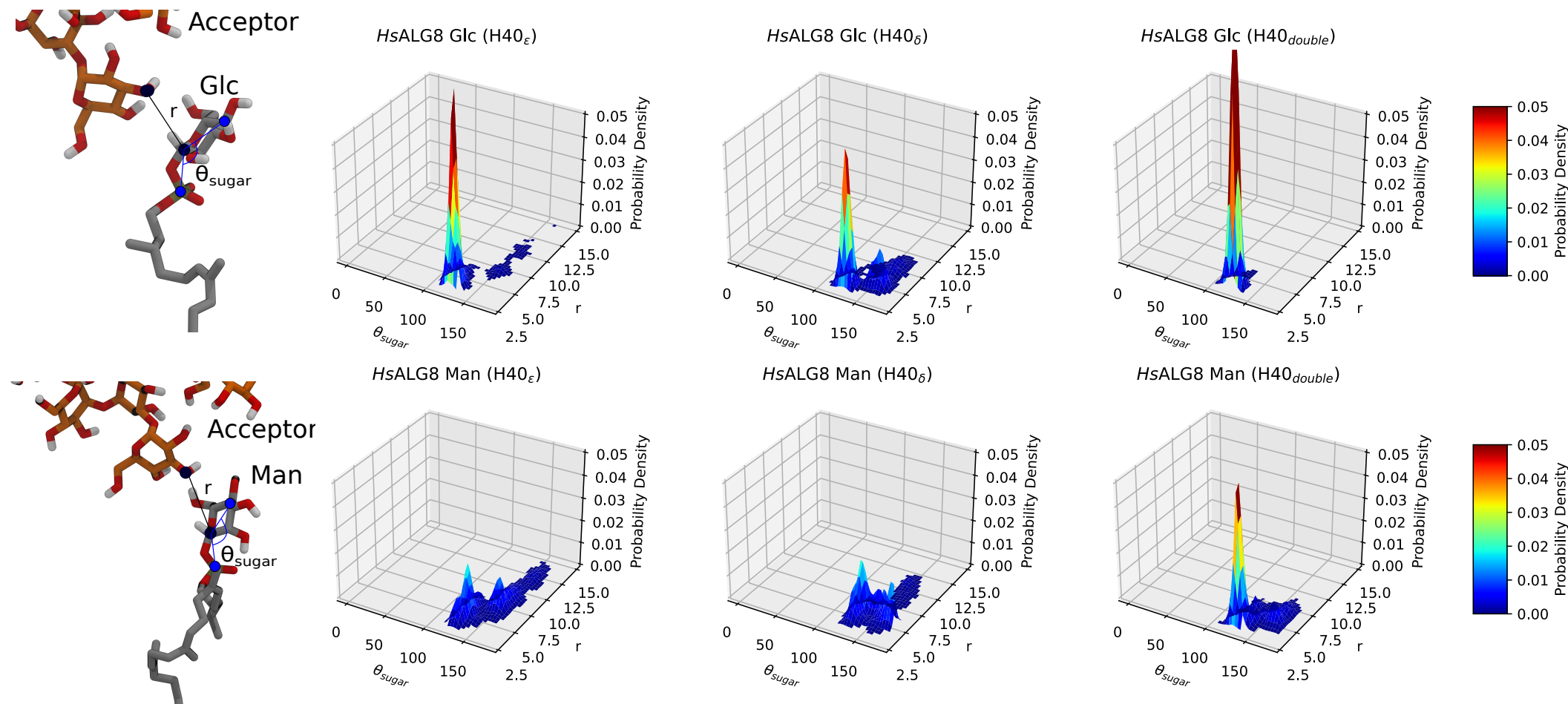
Chapter 2

Pretransfer:

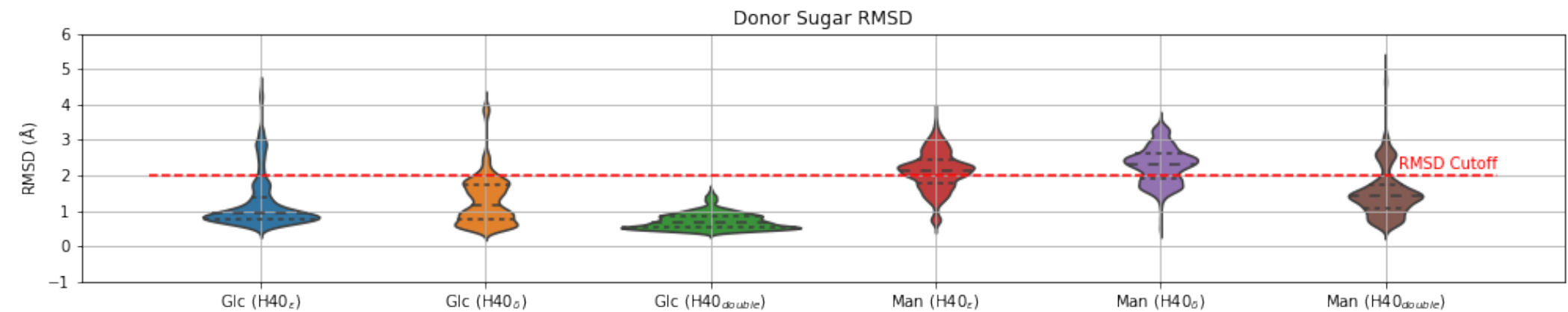
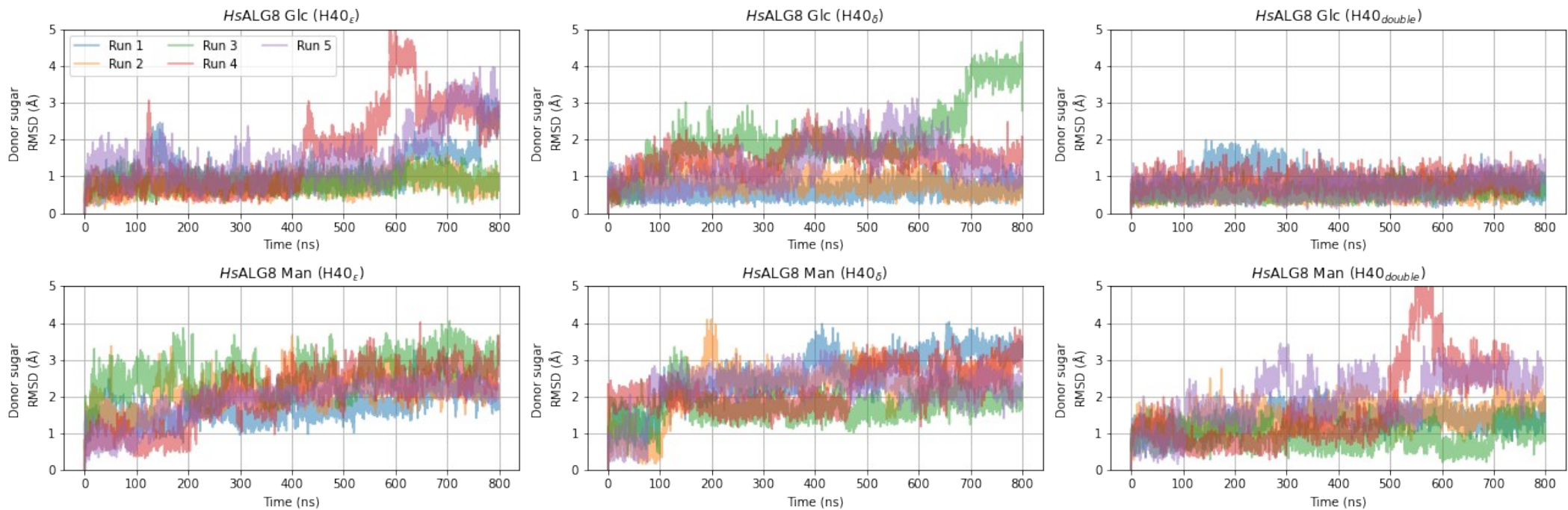
Why is Mannose not transferred?

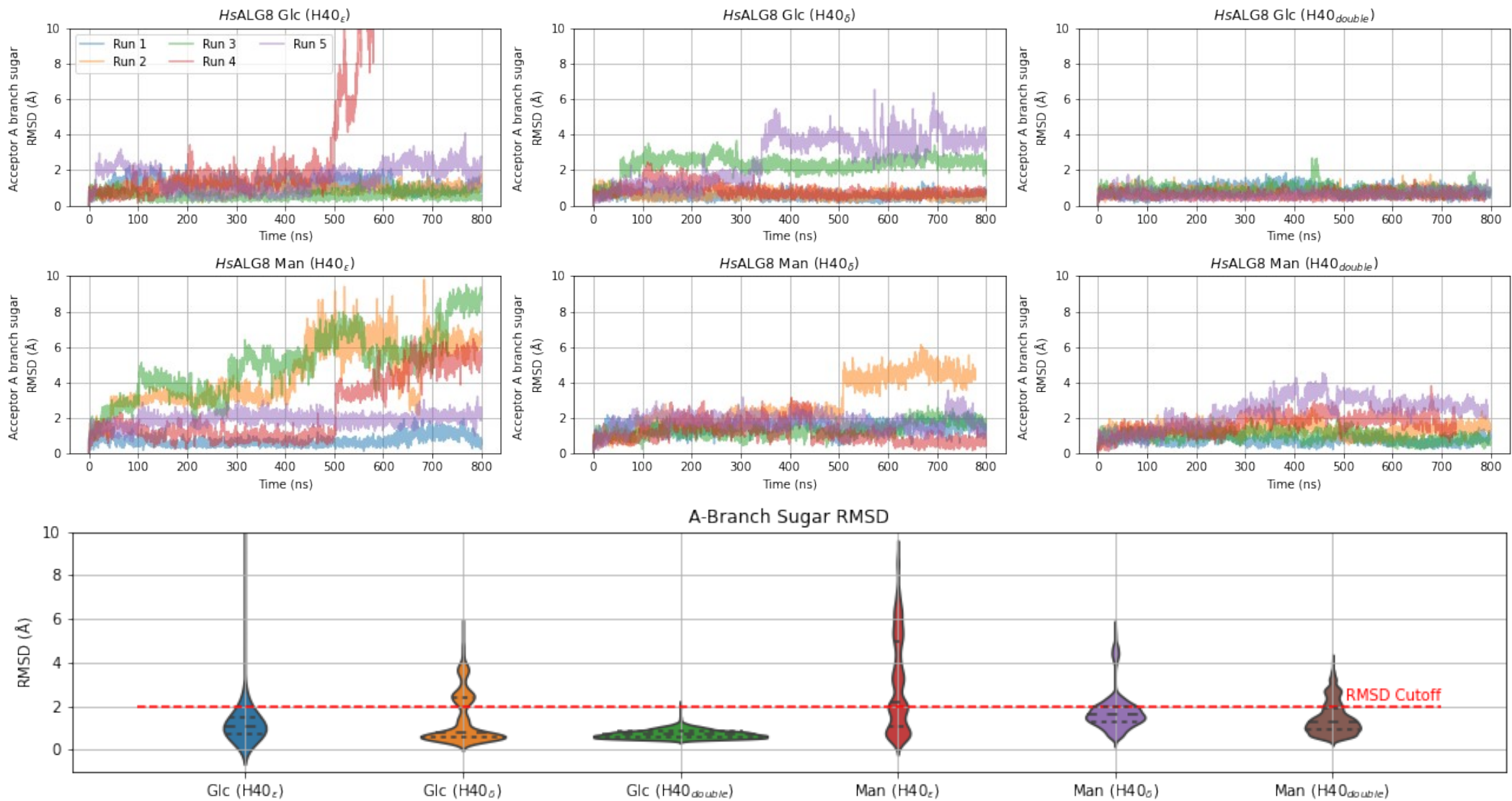
Active Geometry





Mannose donor fails to form a “bent” conformer because of the C2-hydroxyl hydrogen bond with the phosphate group

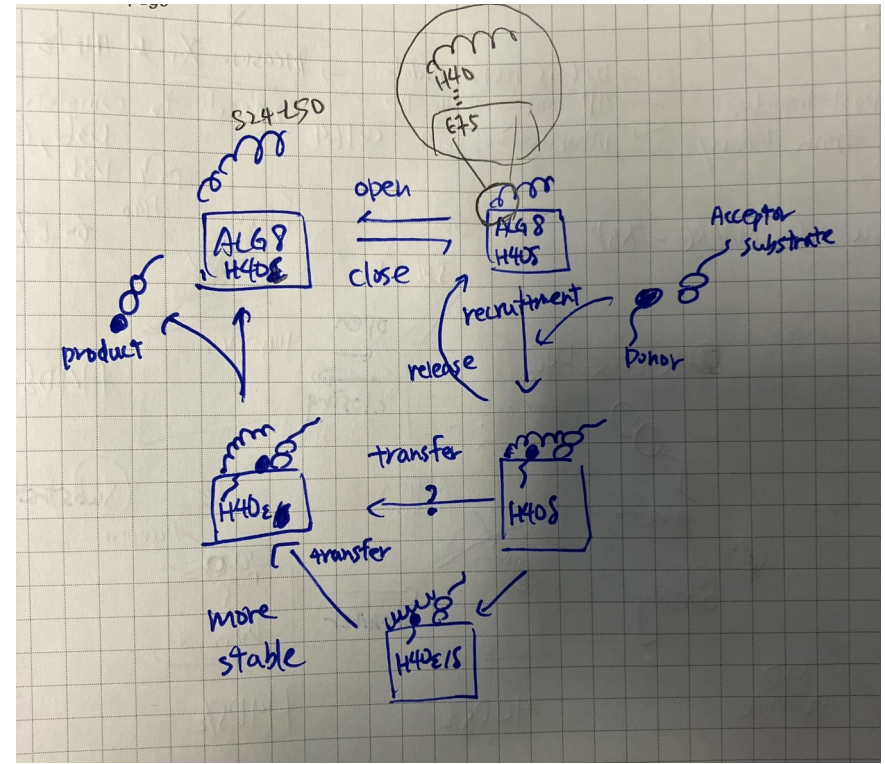




Summary of Chapter 2

- Simulation shows high selectivity of Glc over Man. Potentially because of the conformational limit of the Man-donor.
- Strong correlation between the H40 protonation state and the substrate stability:
 - $D36_{H40\epsilon/\delta} \gg D36_{H40\delta} > D36_{H40\epsilon}$
- Note that the Hbond between H40 and E75 is well-resolved in the EM structure.

Hypothesis



H40 protonation state regulates the substrate recruitment and release:

- D36_{H40ε} induces an open ALG8 conformation. → Product release
- D36_{H40δ} induces the tightening of the active site geometry by hbond with E75.
→ substrate recruitment and donor selection.
- D36_{H40ε/δ} makes the active geometry super stable. → catalysis